ENGINEERING MATHEMATICS-II

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FOURIER TRANSFORMS

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INTRODUCTION

A transform is a is a mapping between two functions.
Fourier Transform is a type of Integral transform.
It converts functions in time domain to frequency domain.
It was discovered by **Jean Baptiste Joseph Fourier** while seeking solution of Heat Equation.

Integral Transform:

An **integral transform** maps a function from its original function space into another function space using integration, where some of the properties of the original function might be more easilycharacterised and manipulated than in the ori ginal function space.

The transformed function can be mapped back to the original function space using the **inverse transform.**

Integral Transform:

The integral transform F(s) of a function f(x) defined by

$$F(s) = \int_a^b f(x)\, K(s, x)\, dx$$

provided the integral is convergent and

K(s,x) is called the kernel of the transform.

Depending upon (a, b) and K(s, x) we get different transforms.

When (a, b) =(0, ∞) and K(s, x) = $\mathbf{e^{-sx}}$ we get Laplace Transforms

When (a, b) =(-∞,∞)and K(s, x) = $\mathbf{e^{isx}}$ we get Fourier Transforms

The function f(x) is called the **Inverse transform of F(s).**

INTRODUCTION

Conversion to Frequency domain:

The Fourier Series showed us how to rewrite any periodic function into a sum of sinusoids.

The Fourier Transform is the extension of this idea to non-periodic (aperiodic) functions.

The Fourier Transform is a tool that breaks a function (waveform or signal) into an alternate representation, characterized by sines and cosines.

Dirichlet's Conditions:

The necessary conditions for existence of Fourier Transform are:

The function f(x) should be absolutely integrable i.e.,

$$\int_{-\infty}^{\infty} |f(x)|\, dx < \infty$$

The function must have finite number of maxima and minima.

The function must have finite number of finite discontinuities.

DEFINITION: FOURIER TRANSFORMATION

Definition: The Fourier Transform of *f(t)* or complex Fourier transform of $f(t)$ is denoted as $F(\omega)$ and defined as

$$\mathcal{F}[f(t)] = F(\omega) = \int_{-\infty}^{\infty} f(t)\,.\,e^{-i\omega t}\,dt$$

DEFINITION: FOURIER INVERSE TRANSFORMATION

The inverse Fourier transform of $f(t)$ is given by,

$$\mathcal{F}^{-1}\{\mathcal{F}[f(t)]\} = \mathcal{F}^{-1}\{F(\omega)\}$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega)\ e^{i\omega t}\, d\omega$$

SPECIAL CASE OF FOURIER TRANSFORMATION

By the definition of bilateral Laplace transform

$$L[f(t)] = F(s) = \int_{-\infty}^{\infty} e^{-st} f(t)\, dt$$

where $s = \sigma + i\omega$.

If we set $\sigma = 0$, we obtain

$$F(\omega) = \int_{-\infty}^{\infty} e^{-i\omega t} f(t)\, dt$$

which is the Fourier transform of $f(t)$.

FOURIER TRANSFORMATION OF SOME STANDARD FUNCTIONS

1. $f(t) = e^{iat}$

By frequency shifting property, we have

$$\mathcal{F}\left[e^{iat} f(t)\right] = F(w - a)$$

Also $\mathcal{F}[1] = 2\pi\delta(\omega)$

$$\therefore \mathcal{F}\left[1.e^{iat}\right] = \mathcal{F}\left[e^{iat}\right] = 2\pi\delta(w - a)$$

FOURIER TRANSFORMATION OF SOME STANDARD FUNCTIONS

2. Trigonometric functions, $f(t) = \cos at$

$$\mathcal{F}[\cos at] = \mathcal{F}\left[\frac{e^{iat} + e^{-iat}}{2}\right] = \frac{1}{2}\{\mathcal{F}[e^{iat}] + \mathcal{F}[e^{-iat}]\}$$

$$\mathcal{F}[\cos at] = \frac{1}{2}\{2\pi\delta(\omega - a) + 2\pi\delta(\omega + a)\}$$

$$\therefore \mathcal{F}[\cos at] = \pi\delta(\omega - a) + \pi\delta(\omega + a)$$

$$= \pi\{\delta(\omega - a) + \delta(\omega + a)\}$$

FOURIER TRANSFORMATION OF SOME STANDARD FUNCTIONS

3. $f(t) = \sin at$

$$\mathcal{F}[\sin at] = \mathcal{F}\left[\frac{e^{iat} - e^{-iat}}{2i}\right] = \frac{1}{2i}\{\mathcal{F}[e^{iat}] - \mathcal{F}[e^{-iat}]\}$$

$$\mathcal{F}[\sin at] = \frac{1}{2i}\{2\pi\delta(\omega - a) - 2\pi\delta(\omega + a)\}$$

$$= \frac{1}{i}\{\pi\delta(\omega - a) - \pi\delta(\omega + a)\}$$

$$\therefore \mathcal{F}[\sin at] = -i\pi\delta(\omega - a) + i\pi\delta(\omega + a)$$

$$= i\pi\{\delta(\omega + a) - \delta(\omega - a)\}$$

THANK YOU

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